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ACTL2131 Assignment 2

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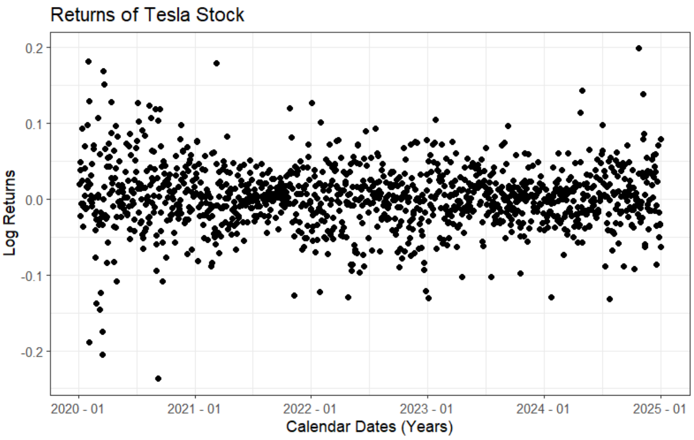
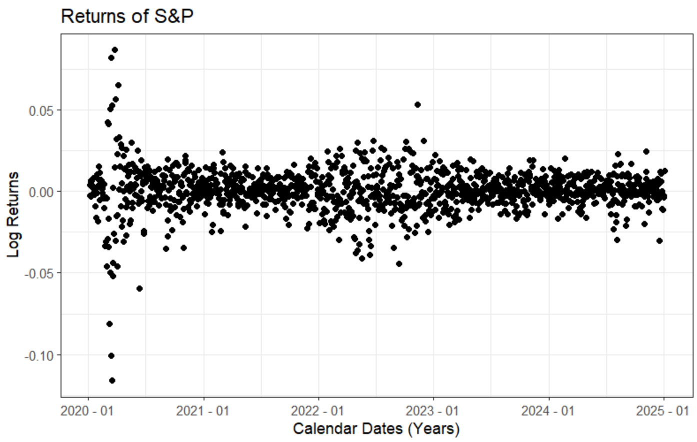
Chosen Stock: Tesla (TSLA) **Comparison Period: (2/1/2020 – 1/1/2025)**

1. Analysis of the Log Returns for Tesla and the S&P 500.





Log return scatter plots for Tesla (left) and the S&P 500 (right) show that returns are generally centred around zero but exhibit high volatility, periodically clustering in the extremes during market downturn, notably in 2020 and 2022 when returns for both assets experienced large positive and negative swings. Additionally, the datasets appear to not follow any seasonal trends. However, Tesla's returns are more visibly more dispersed, covering a larger region of values. This is especially clear from 2020-2021 and 2023-2025, where Tesla returns demonstrate more frequent and extreme fluctuations while the S&P 500 appears mostly clustered around zero apart from occasional spikes .

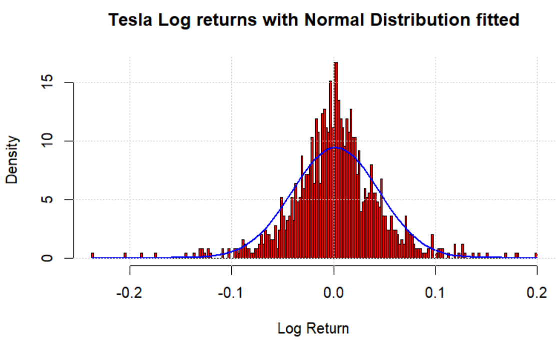
2. Summary Statistics

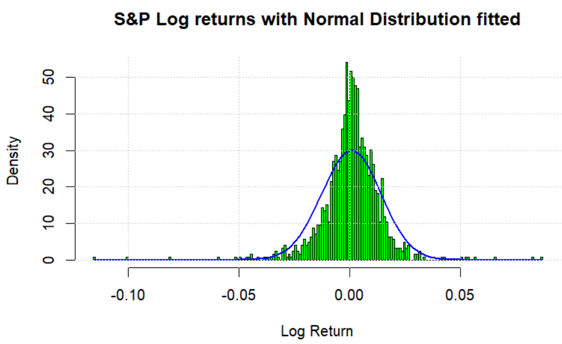
Statistic	Tesla	S&P ETF
Mean	0.0021	0.0005
Variance	0.0018	0.0002
Skewness	-0.1086	-0.8114
Kurtosis	6.1869	15.3225

The summary statistics support the earlier observations. The mean returns for both datasets are marginally greater than 0 and Tesla stocks exhibit a variance 10 times greater than the S&P 500. Both datasets are negatively skewed indicating that the datasets have longer left tails corresponding with a higher likelihood for extreme negative returns. Interestingly, the S&P's skewness is greater than Tesla's and is due to the extreme outliers created by market downturns having a comparatively more pronounced effect on the S&P's lower variance data. Both distributions have Pearson kurtosis values greater than 3, suggesting heavier tails than a normally distributed data set and a higher frequency of extreme returns. Once again, the S&P index shows a larger kurtosis as its lower variance makes extreme values stand out more as outliers.

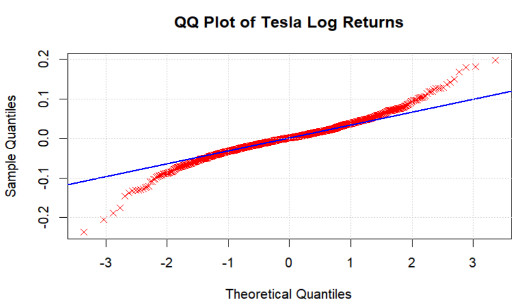
3. Return Histograms

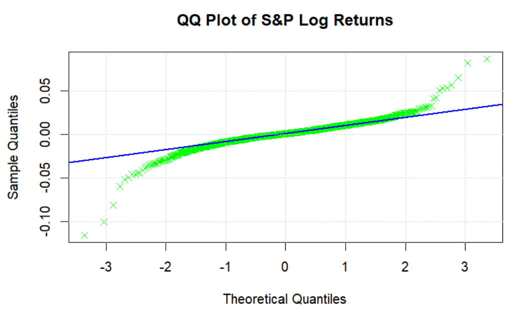
Both datasets, (Tesla left, S&P right) somewhat follow the fitted normal distribution but are more clustered around the mean and as a consequence, are under-represented in the percentiles away from the mean. The normal distribution appears to be a worse fit for the S&P index as its negative skew is more clear and creates a taller peak at the mean.





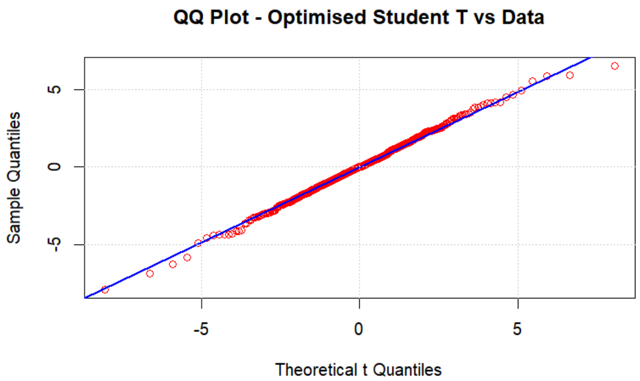
4. The distribution of Tesla and the S&P 500's returns

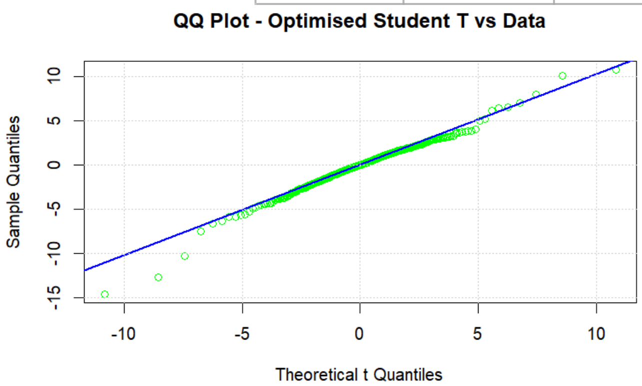




To test whether the two data sets follow normal distributions, QQ-plots are constructed. These QQ-plots (see previous page) demonstrate how the normal distributions provide a reasonable fit for the central quantiles, but fail to capture behaviour outside of this region. Thus, the manager's suspicions are valid and so maximum likelihood estimations are used to find the parameters of their best fitting t-distributions. Initial t-distribution parameters are set as sample mean for location, sample standard deviation for scale and 5 as a starting point for degrees of freedom. This is then optimised yielding the final parameter values in the table to the right.
To test the fit of the data with these t-distributions, QQ-plots are created.

Variable	Tesla	S&P ETF
Location (μ)	0.0019	0.0010
Scale (σ)	0.03018	0.0080
Degrees of Freedom	3.8027	3.0357





These QQ-plots demonstrate how the fitted t-distributions are comparatively better representations of the data. This is particularly evident for Tesla returns, as the tails of S&P datasets are still underrepresented. This likely occurs as outlier returns caused by financial events will appear more pronounced on the S&P datasets as it has a lower variance than Tesla.

To further assess the fit of their t-distributions, a chi-squared goodness of fit test is performed. This test assumes that each observation is independent, although this is not true for financial data limiting the validity of the results. Regardless, we proceed with the test as its large sample size allows the data to be grouped in a large number of quantile bins. A hypothesis test is then conducted with the null hypothesis that each dataset follows their respective t-distribution and with a test statistic created by comparing the frequency of observed data points with their expected results from the theoretical distributions. This yielded p-values of 0.72 and 0.18 for Tesla and S&P returns respectively. Crucially, these values are greater than the significance level meaning that t-distributions are good fits for the log returns of Tesla and S&P stock. Moreover, these results confirm our observation that the t-distribution is a better fit for Tesla's returns as its p-value is greater than the S&P p-value.

Q5. Comparison of annualised returns between S&P and Tesla stock.
A two sample t-test is conducted to analyse the manager's hypothesis that the annualised mean return for the two stocks are equal. Taking the null hypothesis to be that the annualised returns are the same and an alternate hypothesis that the two means differ, a p-value of 0.22 is returned, greater than the significance level of 0.05. Hence, there is insufficient evidence to reject the null hypothesis. However, earlier operations demonstrated significantly different sample means. Thus, we will test the validity of this test.

The two sample t-test, assumes that the population variances are equal and that the samples are independent of each other. In this scenario, observations are not independent as Tesla is a component of the S&P index. Additionally, the assumption of equal variances may be tested through a difference in variance f-test. This test yielded a near 0 p-value, resulting in a rejection of the null hypothesis that the population variances were equal. Thus, given that the two assumptions made by the test are invalid, the result from the two sample t-test is limited and should be interpreted with healthy scepticism. .

Q6. Magnitude of the annualised return of the Tesla Stock
To test the manager's hypothesis that the annualised return of Tesla stock is 7%, we may perform a hypothesis test taking the null hypothesis to be that the mean annualised return is 7% and an alternate hypothesis that the mean annualised return is not 7%. To perform this test, we can create a test statistic through applying the central limit theorem to the sample mean. While the data has over 1000 samples, these samples are not independent of each other. Regardless of the invalidity, the testing process proceeds creating a test statistic using the central limit theorem to assume that the sample mean follows a standard normal distribution. The hypothesis test will then yield a p-value of 0.06, greater than the significance level. Thus, the null hypothesis is not rejected despite the sample mean of the annualised returns for Tesla stock being 52%, far greater than the hypothesised value of 7%. This is explained by the Tesla stock's high variance. While the sample mean is more than 7 times larger than the hypothesised value, the high variance in Tesla's returns means the data remains plausible under the null hypothesis and so the observed sample is not convincing enough to reject the null hypothesis.